Bayesian Optimisation of a Bio-Based Biodegradable Polyester Thermoset

29th March 2025

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Abstract

Glass reinforced plastic composites are ubiquitous in the wind turbine manufacturing industry, and are commonly manufactured using general-purpose unsaturated polyester resin thermosets. Unfortunately, these resins are derived from oil-derived aromatic monomers making them both unsustainably sourced and hard to recycle. Bio-based and biodegradable alternatives exist but have not seen widespread commercial success; this paper seeks to address the question of whether bio-based branching thermoset polyesters can be optimised using Bayesian optimisation such that desirable properties are improved upon. This is achieved by identifying a bio-based and biodegradable polymer, poly(glycerol citrate itaconate), identifying a desirable objective variable and some tunable predictor variables, and finally using Bayesian optimisation to sample-efficiently identify well-performing samples. The most salient findings included the Bayesian optimisation routines successfully optimising the proxy variable in a sample efficient fashion, improving upon common deterministic techniques such as grid sampling.

1 Introduction

In 1987, the concept of 'sustainable development' was defined by the United Nations (UN) Brundtland Commission as a development which seeks to "meet the needs and aspirations of the present without compromising the ability to meet those of the future" [25]. Wind turbine manufacturing is an industry shaped by this ideal, with its end-products contributing towards the UN's 13th sustainable development goal (SDG) which urges "urgent action to combat climate change and its impacts" [23]. Unfortunately, wind turbine blades are commonly manufactured from a composite comprised of thermosetting polymer matrix and reinforcing material, these strong thermoset matrices are chemically cross-linked which makes them difficult to reprocess or remould [14]. Given this resistance to recycling and a relatively common rate of replacement (ca. every fifteen years in China [8]) this industry generates a large volume of practically unrecyclable thermoset waste which contravenes the UN's 12th SDG to "ensure sustainable consumption and production" [23]. In addressing the industry's sustainability shortcoming identified above, the literature forks into two groups: firstly, papers which focus upon improving the recyclability of contemporary materials, whether by thermal, chemical, mechanical or high-voltage fragmentation methods [24, 5, 9]. Secondly, there are papers aimed at the development of properties in future replacement materials that would facilitate recycling [13], such as introduction of chemically cleavable bonds into the thermoset polymers used [21]. This paper's research sought to ascertain whether Bayesian optimisation could be used to sample efficiently improve a desirable property in a novel biodegradable bio-based thermoset.

The composite matrix used for wind turbine blades is comprised of a moulding thermoset, with the most prevalently manufactured pre-thermoset form being general-purpose unsaturated polyester resin (GP-UPR) [6]. These GP-UPRs are based on a mixture of reactive diluent (e.g. styrene)

and linear aromatic unsaturated polyester (e.g. poly(propylene glycol maleate phthalate)), which are cured chemically or thermally to generate the final thermosetting matrix [1]. Although polyesters are known to exhibit biodegradability due to their labile ester bonds being prone to both biotic and abiotic hydrolysis [18], it is the case that polyesters with aromatic moieties are more resistant than aliphatic counterparts to biodegradation [16]. Consequently, thermosets derived from GP-UPRs synthesised from the aromatic monomer phthalic acid [1] can be reasonably expected to exhibit a high resistance to biodegradation; which was part of their original allure as a material. An archetypical example of an aliphatic polyester would be poly(glycerol sebacate), which uses the monomers glycerol (triol) and sebacic acid (dicarboxylic acid) to form a branching polyester thermoset on thermal curing via polyesterification [27]. Given their natural biodegradability, this research chose to focus upon aliphatic polyesters.

Another concern associated with GP-UPRs with respect to the UN's 12th SDG is that of a material's provenance; if the underlying chemistry of a polymer is petrochemical in origin, then any carbon liberated from the material at end-of-life (whether by combustion for energy or biodegradation in landfill) will release fossil carbon from the long-term carbon cycle into the short-term carbon cycle that exists above the endogenic reservoir, contributing via an anthropogenic route towards global climate change. Since the aforementioned GP-UPRs are synthesised from the oil-derived feedstock monomers propylene glycol, maleic acid, and phthalic acid, the resultant thermosets contain endogenic (fossil-derived) carbon [12]. This research avoided these feedstocks and centred on a bio-based alternative, with inspiration drawn from a chemical system presented in the literature by Alberts, where a combination of glycerol and citric acid were cured to form a bio-based branched aliphatic polyester thermoset [2]. We build upon this research by adopting the chemical system and adapting it slightly to include a third monomer, itaconic acid, which provides a second carboxylic acid with a lower degree of functionality (dicarboxylic acid). Notably, Alberts found that physical properties of the end thermoset poly(glycerol citrate) could be tuned by modifying the monomeric stoichiometries of the prepolymer resin synthesised [2]. Given this, we identified mass loss as a variable of interest during the synthesis of poly(glycerol citrate itaconate) as it was a proxy for polymerised content within a sample, and sought to optimise this by tuning the stoichiometries of the prepolymer resins mixed.

Poly(glycerol citrate itaconate) is formed from its monomers by polyesterification, a process which is dependent upon multiple variables such as temperature, stoichiometric ratios, and additive presence [2]. Additionally, reaction times for rigid solid thermoset materials are on the order of weeks when uncatalyzed [22] (catalysis commonly involves strong fuming acids which corrode equipment and are best avoided [19]). Given the high dimensionality and temporally expensive function evaluation exhibited by this family of polyesters, a sample efficient global optimisation technique by the name of Bayesian optimisation was elected as a suitable means of optimisation. Alternative global optimisation techniques were considered but ruled out; the hill-climbing class gradient descent algorithm was reliant on derivatives of the objective function [20], which would have been uncertain and practically difficult to obtain given the noisy real-world nature of the black-box function evaluated. The population class evolutionary and particle swarm optimisation algorithms were suitable but avoided in favour of the more sample efficient surrogate class of algorithms. Surrogate class algorithms are variously formulated, but the three major forms are distinguished by the surrogate models they leverage- polynomial response surface model , kriging (aka gaussian process re-

gression), radial basis function [17]. Kriging is preferable to response surface modelling since it makes no assumption of polynomial order, avoiding a priori bias in surrogate models generated. Furthermore, radial basis function based surrogate models provide no estimates of uncertainty around the fitted function; functions and associated uncertainties derived using kriging provides the means by which bayesian optimisation effectively balances exploitation and exploration of the parameter space. This research therewith saw the development of a part-computational and part-laboratory based workflow for the optimisation of mass loss in poly(glycerol citrate itaconate) leveraging Bayesian optimisation predicated on kriging.

Originally coined in the 1970s by Mockus et al. [15], Bayesian optimisation was later developed further in the 1990s by Jones et al. into a technique called the efficient global optimisation algorithm [11]- this name and technique was thereafter adopted and popularised by the DoE community. More recently, the original name has seen a revival as the technique and a plethora of adaptations have been re-discovered and implemented in a range of contemporary communities including robotics [7], bioengineering [3], and machine learning [10]. In materials science, Bayesian optimisation has been used to predict crystalline structures [26], carry out sample-efficient optimisation of density functional theory simulations [4], and multi-objective optimisation of stability and ion-conductivity in materials

2 Method

This chapter will elucidate the various practicalities associated with scientifically researching the research questions posed in the introduction.

$$y = f(x_1, x_2, x_3) \tag{1}$$

$$f : \mathbb{R}^n \to \mathbb{R} \tag{2}$$

$$f(\mathbf{x}) : \mathbf{x} = (x_n, x_{n+1}, x_{n+2}) \tag{3}$$

$$\begin{aligned} &R - COOH + HO - R' \to \\ &R - COO - R' + H - O - H \end{aligned} \tag{4}$$

$$\begin{aligned} &RCOOH + HOR' \to \\ &\qquad RCOOR' + HOH \end{aligned} \tag{5}$$

2.1 Background on Chemical System Researched

The chemical system was a branching polyester by the name of poly(glycerol itaconate citrate). It is based on three simple monomers all of which are readily available bio-based feedstocks- glycerol, itaconic acid, and citric acid. When heated for an extended period of time these polyesterify, which is to say, undergo esterification in a chained fashion. Due to the multi-functional nature of these chemicals this generates branching polyesters, which can cross-link with themselves, creating a thermoset product.

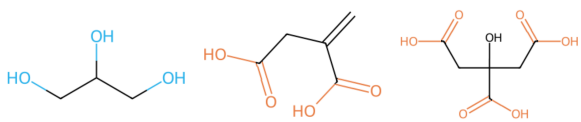
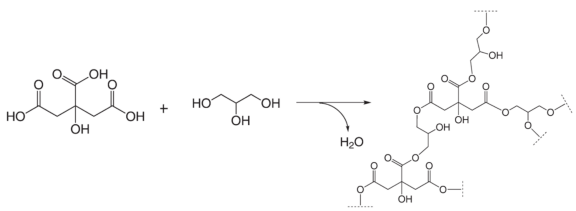


Figure 1: *Skeletal diagrams of chemicals studied. Left: Glycerol. Centre: Itaconic acid. Right: Citric acid. Functional groups considered likely to be taking part in the esterification are coloured: Blue for hydroxyl groups, orange for carboxyl groups.*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2.2 Target Variable Selected

Given the proof-of-concept nature of this project we elected to use a proxy variable for strength. This would be mass loss; the mildly naive rationale behind this was as follows: For each ester link generated during the polyesterification process a water molecule is generated. If the heating temperature is above 100oC, which is above the boiling point of water, then water should naturally evaporate over the course of the polyesterification reaction. This would generate a mass loss between the start mass of the various monomers, and the end mass of the polyesterified thermoset product. We hence make two basic assumptions: firstly, mass loss is directly proportional to the number of ester linkages generated. Secondly, we assume that the number of ester linkages generated should be proportional to the number of successful cross-links within the final 3D structure of the thermoset. Thirdly and finally, we assume that the number of cross-links generated in the final 3D structure of the thermoset is proportional to the tensile strength exhibited by the thermoset.

It is therewith trivial to discover the value of this proxy variable- one simply weighs the mass of monomers used beforehand and the mass of the resultant thermoset after the polyesterification (See equation ??). Important is the problem of how to call an end to the polyesterification process. This ended up taking the form of an arbitrary gradient, beyond which the polyesterifiation would be terminated.

2.3 Predictor Variables, How to Tune Them Practically, and the Parameter Space

Given that poly(glycerol itaconate citrate) is comprised of three monomeric components, it becamce clear that the chemical stoichiometry would be the primary means by which properties could be tuned. This meant building a balancing system for the various compositions possible, and placing some bounds on the parameter space. (Provide the graphics used in the first year report here...)

2.4 Bayesian Optimisation of the Variables and How it was Implemented Computationally and Practically

Given the target variable and predictor variables, a prior could be built over the parameter space. This was achieved using various forms of deterministic random sampling techniques. Once the prior was generated, this could be leveraged to carry out Bayesian optimisation in various forms. Description of the various stepping methods et cetera.

2.5 Benchmark Techniques

To prove the sample efficient nature of Bayesian optimisation we used a number of deterministic grid sampling techniques. These were grid, quasirandom, and pseudorandom, and used the entire sample limit we had presribed. Naturally, grid was the most representative of the parameter space, and found a good coupld of points. Whilst, quasirandom under performed relatively, and pseudorandom failed miserably... Some benchmarks have been developed, these are presented in figure ??.

3 Results

This results section is dedicated to presenting and contrasting the various optimisations undertaken during the research. It opens by presenting the space-filling deterministic sampling routines, which act as simple benchmarks for the progress of the Bayesian techniques. It then moves onto the various Bayesian techniques trialled.

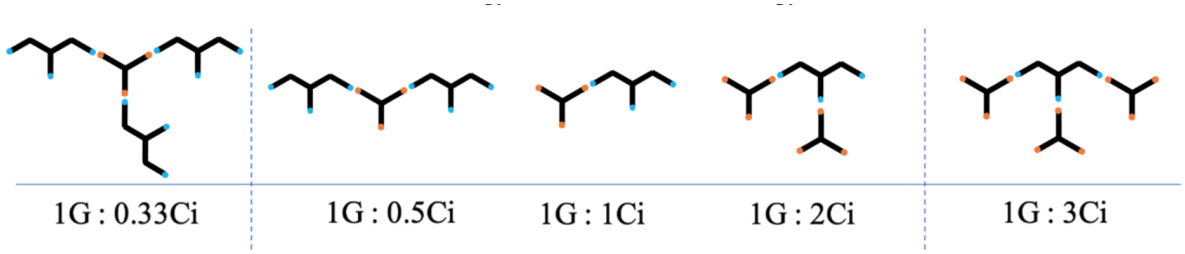

Figure 3: *Illustration of stoichiometric ratios between glycerol (G) and citric acid (Ci) and possible repeat units.*

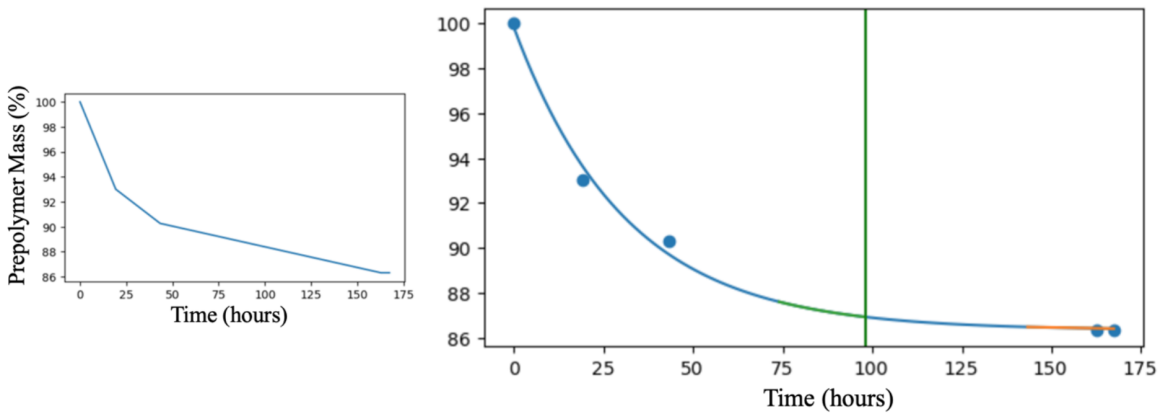

Figure 4: *Left: Prepolymer mass loss over time, data points connected by straight lines. Right: Prepolymer mass loss over time, data points scattered, and blue exponential curve fitted. Green vertical line shows time at which target gradient reached, and straight green tangent line exponential shows linear fit for last 24 hours of data. Orange line gives the latest 24hr tangent available at the time of graph drawing.*

3.1 Benchmark Deterministic Space-Filling Sampling

Insert table of results here.

3.2 Bayesian Techniques

3.3 Comparisons of Optimisation Progressions

The optimisation progressions are presented to show the progress of optimisations and how they break the simple benchmarks laid out earlier.

Insert table of results here.

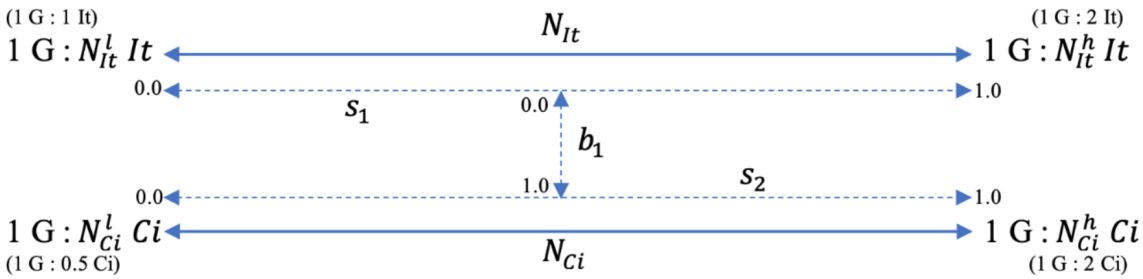

Figure 5: *Visualisation of 3D optimisation parameter space for the P(GCiIt) problem.*

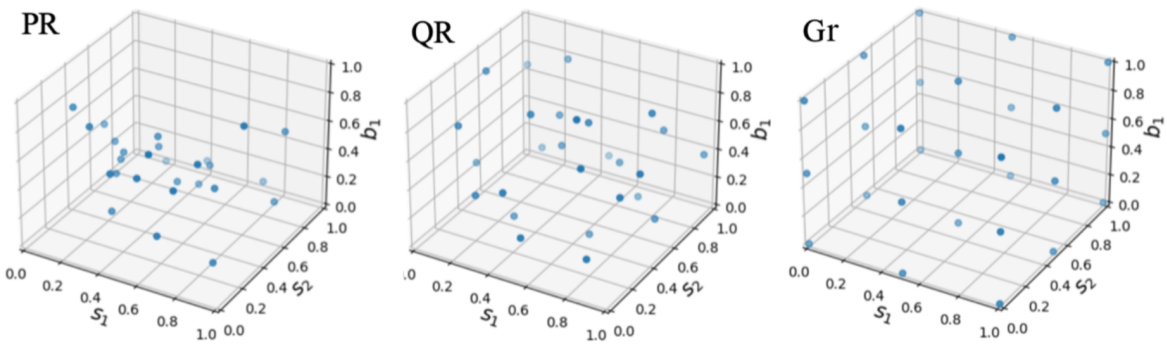

Figure 6: *Comparison of space-filling sampling techniques and distribution of 27 samples across the entire 3D parameter space of the P(GCiIt) problem. G: Grid sampling technique, QR: Quasirandom sampling technique, PR: Pseudorandom sampling technique.*

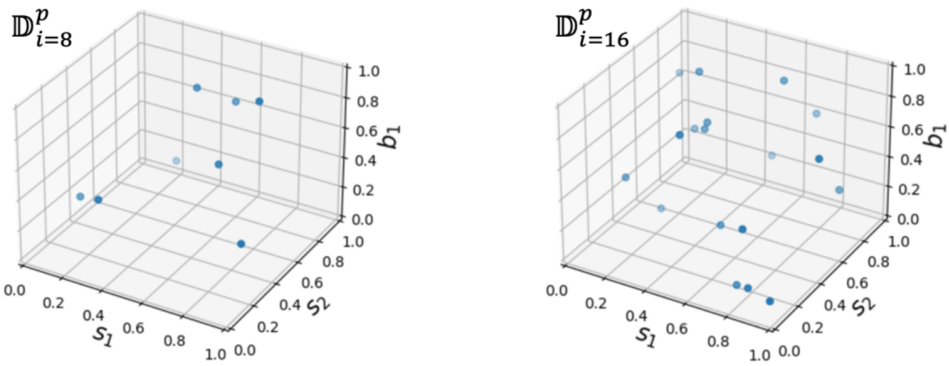

Figure 7: *Visualisation of the two representative prior datasets quasirandomly across the 3D parameter space of the P(GCiIt) system. : Eight sample prior dataset. : Sixteen sample prior dataset.*

4 Discussion

The discussion comprises all of the notable findings (set in context) drawn from the datasets objectively presented in the results section. **Prior-Forming Dataset Size Effect** ahbx

Acquisition Function Effect ahbx

Bayesian Optimisation Sample Efficiency ahbx

Stochastic Sampling Problem ahbx

Global Optimum Estimation ahbx

Optimum Stoichiometry Derivation ahbx

5 Conclusion

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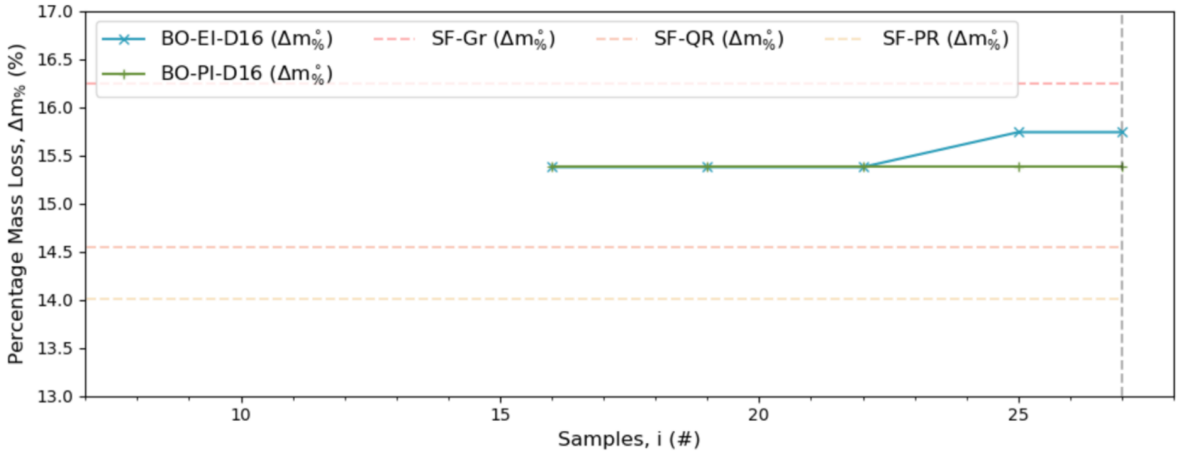

Figure 8: *Graph describing optimisation progression with respect to best global optimum discovered at any given iteration for BO-EI-D16 and BO-PI-D16. Black dashed vertical line denotes ???. Crosses or pluses distributed along solid lines denote the ??? at the end of an iteration. SF-Gr/QR/PR: Space-filling Grid/Quasirandom/Pseudorandom techniques. BO: Bayesian optimisation method. PI: Probability of improvement acquisition function. EI: Expected improvement acquisition function. D16: 16-sample prior-initialising dataset.*

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6 Bibliography

[1] A Ahamad et al. "Synthesis of unsaturated polyester resin - effect of anhydride composition". In: *Designed Monomers and Polymers* 4.3 (2001), pp. 261–268.

[2] Albert Alberts et al. "Plantics-GX: a biodegradable and cost-effective thermoset plastic that is 100% plant-based". In: *Faraday Discussions* 202.0 (2017), pp. 111–120.

[3] J Bader et al. "Improving extracellular vesicles production through a Bayesian optimization-based experimental design". In: *European Journal of Pharmaceutics and Biopharmaceutics* 182 (2023), pp. 103–114.

[4] P Balachandran et al. "Adaptive strategies for materials design using uncertainties". In: *Scientific Reports* 6 (2016).

[5] J Beauson et al. "Recycling of shredded composites from wind turbine blades in new thermoset polymer composites". In: *Composites: Part A* 90 (2016), pp. 390–399.

[6] S Bobade et al. "Bio-based thermosetting resins for future generation: a review". In: *Polymer-Plastics Technology and Engineering* 55:17 (2016), pp. 1836–1896.

[7] R Calandra et al. "Bayesian optimization for learning gaits under uncertainty". In: *Annals of Mathematics and Artificial Intelligence* 76 (2016), pp. 5–23.

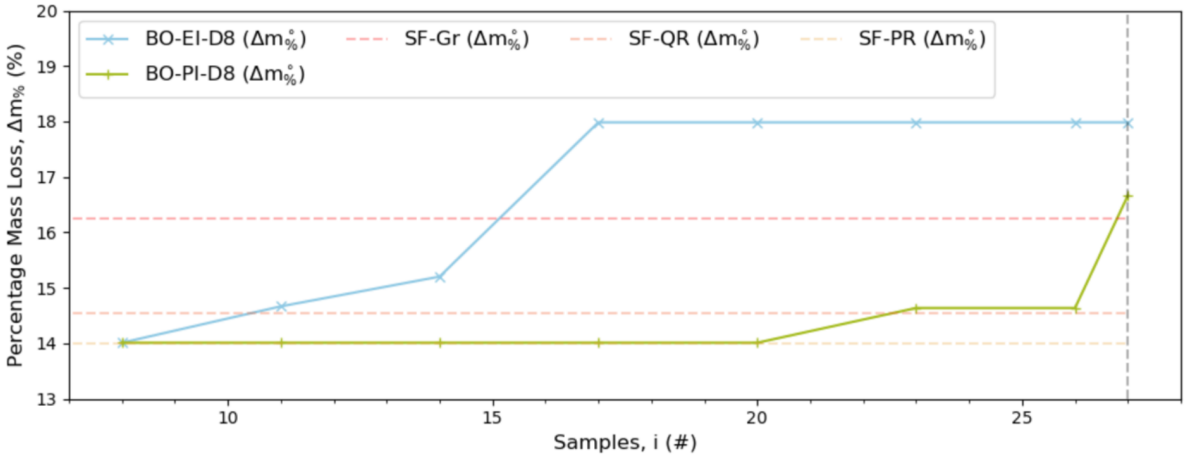

Figure 9: *Graph describing optimisation progression with respect to best global optimum ??? discovered at any given iteration for BO-EI-D8 and BO-PI-D8. Black dashed vertical line denotes ???. Crosses or pluses distributed along solid lines denote the ??? at the end of an iteration. SF-Gr/QR/PR: Space-filling Grid/Quasirandom/Pseudorandom techniques. BO: Bayesian optimisation method. PI: Probability of improvement acquisition function. EI: Expected improvement acquisition function. D8: 8-sample prior-initialising dataset.*

[8] J Chen et al. "Recycling and reuse of composite materials for wind turbine blades: An overview". In: *Reinforced Plastics and Composites* 38 (2019), pp. 567–577.

[9] R Chen et al. "Comparative pyrolysis characteristics of representative commercial thermosetting plastic waste in inert and oxygenous atmosphere". In: *Fuel* 246 (2019), pp. 212–221.

[10] H Cho. "Basic enhancement strategies when using Bayesian optimization for hyperparameter tuning of deep neural networks". In: *IEEE Access* 8 (2020), pp. 52588–52608.

[11] D Jones et al. "Efficient global optimization of expensive black-box functions". In: *Journal of Global Optimization* 13 (1998), pp. 455–492.

[12] P Levi et al. "Mapping global flows of chemicals: from fossil fuel feedstocks to chemical products". In: *Environmental Science and Technology* 52 (2018), pp. 1725–1734.

[13] Y Liu et al. "Closed-loop chemical recyling of thermosetting polymers and their applications: a review". In: *Green Chemistry* 24 (2022), pp. 5691–5708.

[14] S Ma et al. "Degradable thermosets based on labile bonds or linkages: A review". In: *Progress in Polymer Science* 76 (2017), pp. 65–110.

[15] J Mockus et al. "The application of bayesian methods for seeking the extremum". In: *Towards Global Optimisation* 2 (1978), pp. 117–128.

[16] R Müller et al. "Biodegradation of polyesters containing aromatic constituents". In: *Biotechnology* 86 (2001), pp. 87–95.

[17] O Roeva. *Real-world Applications of Genetic Algorithms*. InTech, 2012. ISBN: 978-953-51-0146-8.

[18] S Sharma et al. *A handbook of applied biopolymer technology*. RSC Green Chemistry Series. Cambridge: RSC Publishing, 2011.

[19] K Shukla. "A study on the synthesis of various polyesters from glycerol". In: *Journal of Polymer Research* 29.373 (2022).

[20] J Stork et al. "A new taxonomy of global optimization algorithms". In: *Natural Computing* 21 (2022), pp. 219–242.

[21] G Tesoro et al. "Reversible Crosslinking in Epoxy Resins. I. Feasibility Studies". In: *Applied Polymer Science* 39 (1990), pp. 1425–1437.

[22] J Trenkel-Amoroso. "Synthesis, degradation and practical applications of a glycerol/citric acid condensation polymer". PhD thesis. Oregon State University, 2008.

[23] UN. *The 2030 Agenda for Sustainable Development*. Tech. rep. A/RES/70/1. United Nations, 2015.

[24] S Utekar et al. "Comprehensive study of recylcing of thermosetting polymer composites - Driving force, challenges and methods". In: *Composites Part B* 207 (2020).

[25] WCED. *Our Common Future*. Tech. rep. 019282080X. The United Nation's World Commission on Environment and Development, 1987.

[26] T Yamashita et al. "Crystal structure prediction accelerated by Bayesian optimization". In: *Physical Review Materials* 2 (2018).

[27] H Zhang et al. "Recent advances in glycerol polymers: chemistry and biomedical applications". In: *Macromolecular Rapid Communications* 35 (2014), pp. 1906–1924.

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